

# CM0832 - MT5001 Elements of Set Theory

## 2.5 Orderings

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# Outline

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Introduction

Non-Strict Partial Orderings

Strict Partial Orderings

Linear Orderings

Notable Elements

References

# Outline

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## Introduction

Non-Strict Partial Orderings

Strict Partial Orderings

Linear Orderings

Notable Elements

References

# Preliminaries

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## Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Introduction

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## Description

We can use relations to order **some** or **all** the elements of a set.

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## Examples (some order relations)

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- The words in a dictionary  
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- Academic genealogical descent  
 $aRb$  if  $a$  was the supervisor of the thesis of  $b$ .

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We can use relations to order **some** or **all** the elements of a set.

## Examples (some order relations)

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$aRb$  if  $a$  comes before  $b$  in the dictionary.

- Academic genealogical descent

$aRb$  if  $a$  was the supervisor of the thesis of  $b$ .

- Schedule projects

$aRb$  if  $a$  is a task that must be completed before the task  $b$  begins.

# Introduction

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- What means that a binary relation  $R$  is an ordering relation on a set  $A$ ?

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- Orderings are binary relations.
- What means that a binary relation  $R$  is an ordering relation on a set  $A$ ?
- Non-strict orderings ( $\preceq$ ) and strict orderings ( $\prec$ ) can be interchangeably defined.

# Outline

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Introduction

**Non-Strict Partial Orderings**

Strict Partial Orderings

Linear Orderings

Notable Elements

References

# Antisymmetric Relations

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## Definition [2.]5.1

A binary relation  $R$  in a set  $A$  is **antisymmetric in  $A$**  if for all  $x, y \in A$ ,  $xRy$  and  $yRx$  imply  $x \doteq y$ .

# Antisymmetric Relations

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## Examples

Whiteboard.

# Non-Strict Partial Orderings

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## Definition [2.]5.2

A binary relation  $\preccurlyeq$  in a set  $P$  is a **non-strict partial ordering** if  $\preccurlyeq$  is reflexive, antisymmetric and transitive in  $P$ .

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## Remarks

(a) The adjective “non-strict” means that for all  $a \in P$ ,  $a \preccurlyeq a$ .

# Non-Strict Partial Orderings

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## Remarks

- (a) The adjective “non-strict” means that for all  $a \in P$ ,  $a \preccurlyeq a$ .
- (b) The adjective “partial” means that not every pair of elements of  $P$  needs to be comparable, i.e., there may be  $a, b \in P$ , such that,  $a \not\preccurlyeq b$  and  $b \not\preccurlyeq a$ .

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- (a) The adjective “non-strict” means that for all  $a \in P$ ,  $a \preccurlyeq a$ .
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- (c) The textbook uses the terms “partial ordering” or “ordering” instead of the term “non-strict partial ordering”.

# Non-Strict Partial Orderings

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## Examples

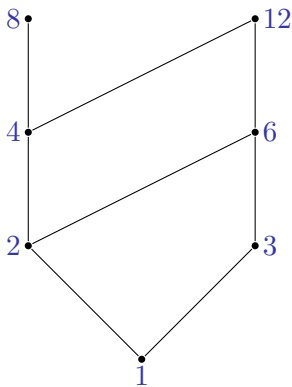
- (a) The relation  $\leq$  is a non-strict partial ordering on the set of all (natural, rational, real) numbers.
- (b) Let  $\subseteq_A$  a binary relation in  $A$  defined by:  $x \subseteq_A y$  if and only if  $x \subseteq y$  and  $x, y \in A$ . The relation  $\subseteq_A$  is a non-strict partial ordering of the set  $A$ .
- (c) The relation  $\text{Id}_A$  is a non-strict partial ordering of  $A$ .

# Hasse Diagrams

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## Example (informal)

Let  $|$  be the relation of divisibility. Hasse diagram for the non-strictly poset  $(\{1, 2, 3, 4, 6, 8, 12\}, |)$ .

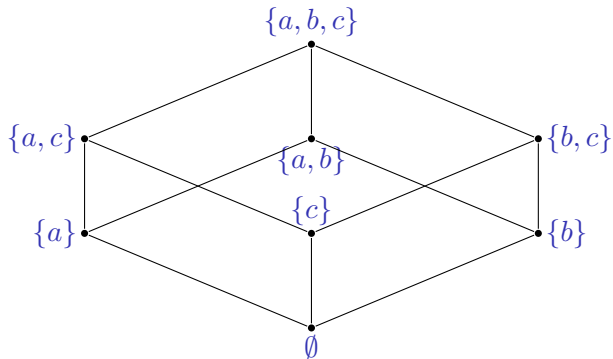


# Hasse Diagrams

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## Example (informal)

Let  $A = \{a, b, c\}$ . Hasse diagram for the non-strictly poset  $(\mathcal{P}(A), \subseteq_{\mathcal{P}(A)})$ .



# Outline

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Introduction

Non-Strict Partial Orderings

**Strict Partial Orderings**

Linear Orderings

Notable Elements

References

# Asymmetric Relations

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## Definition [2.]5.4

A binary relation  $R$  in a set  $A$  is **asymmetric in  $A$**  if for all  $x, y \in A$ ,  $xRy$  implies  $y \not R x$ .

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## Examples

Whiteboard.

# Strict Partial Orderings

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A binary relation  $\prec$  in a set  $P$  is a **strict partial ordering** if  $\prec$  is asymmetric and transitive in  $P$ .

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## Remarks

(a) The adjective “strict” means that for all  $a \in P$ ,  $a \not\prec a$ .

# Strict Partial Orderings

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A binary relation  $\prec$  in a set  $P$  is a **strict partial ordering** if  $\prec$  is asymmetric and transitive in  $P$ .

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- (a) The adjective “strict” means that for all  $a \in P$ ,  $a \not\prec a$ .
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- (c) The textbook uses the term “strict ordering” instead of the term “strict partial ordering”.

# Strict Partial Orderings

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## Examples

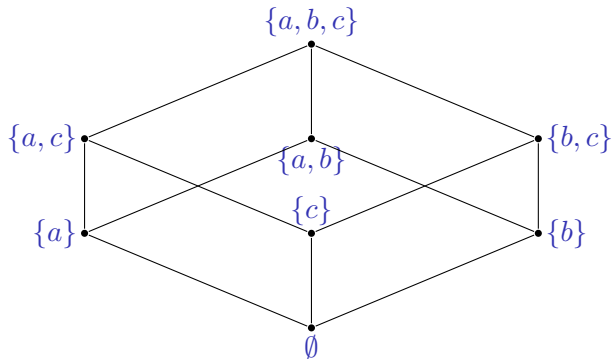
- (a) The relation  $<$  is a strict partial ordering on the set of all (natural, rational, real) numbers.
- (b) Let  $\subset_A$  (proper subset) a binary relation in  $A$  defined by:  $x \subset_A y$  if and only if  $x \subset y$  and  $x, y \in A$ . The relation  $\subseteq_A$  is a non-strict partial ordering of the set  $A$ .

# Hasse Diagrams

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## Example (informal)

Let  $A = \{a, b, c\}$ . Hasse diagram for the strictly poset  $(\mathcal{P}(A), \subset_{\mathcal{P}(A)})$ .



# Partial Orderings

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Theorem [2.]5.6 (relationship between non-strict and strict partial orderings)

(a) Let  $\preceq$  be a non-strict partial ordering on a set  $P$ . The relation

$$x \prec y \stackrel{\text{def}}{=} x \preceq y \text{ and } x \neq y$$

is a strict partial ordering on  $P$ .

# Partial Orderings

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Theorem [2.]5.6 (relationship between non-strict and strict partial orderings)

(a) Let  $\preccurlyeq$  be a non-strict partial ordering on a set  $P$ . The relation

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is a strict partial ordering on  $P$ .

(b) Let  $\prec$  be a strict partial ordering on a set  $P$ . The relation

$$x \preccurlyeq y \stackrel{\text{def}}{=} x \prec y \text{ or } x \dot{=} y$$

is a non-strict partial ordering on  $P$ .

# Outline

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Introduction

Non-Strict Partial Orderings

Strict Partial Orderings

**Linear Orderings**

Notable Elements

References

# Comparable Elements

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## Definition [2.]5.7

- (a) Let  $\preccurlyeq$  be a non-strict partial ordering of  $P$ . Two element  $a, b \in P$  are **comparable** in the ordering  $\preccurlyeq$  if

$$a \preccurlyeq b \quad \text{or} \quad b \preccurlyeq a.$$

# Comparable Elements

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- (b) Let  $\prec$  be a strict partial ordering of  $P$ . Two elements  $a, b \in P$  are **comparable** in the ordering  $\prec$  if

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- (b) Let  $\prec$  be a strict partial ordering of  $P$ . Two elements  $a, b \in P$  are **comparable** in the ordering  $\prec$  if

$$a \dot{=} b \quad \text{or} \quad a \prec b \quad \text{or} \quad b \prec a.$$

- (c) Two elements  $a, b \in P$  are **incomparable** if they are not comparable.

# Linear Orderings

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## Conventions

- (a) The symbol  $\preccurlyeq$  shall denote a non-strict partial order.
- (b) The symbol  $\prec$  shall denote a strict partial order.

# Linear Orderings

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## Definition [2.]5.9

A binary relation  $R$  in a set  $L$  is a **linear ordering** (or **total ordering**) if  $R$  is a partial ordering of  $L$  and any two elements of  $L$  are comparable.

# Linear Orderings

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## Definition [2.]5.9

A binary relation  $R$  in a set  $L$  is a **linear ordering** (or **total ordering**) if  $R$  is a partial ordering of  $L$  and any two elements of  $L$  are comparable.

The pair  $(L, R)$  is a **linearly ordered set**.

# Linear Orderings

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## Examples

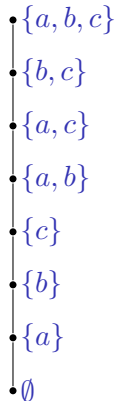
- (a) The relations  $\leq$  and  $<$  are linear orderings on the set of all (natural, rational, real) numbers.
- (b) Neither the relation  $\subseteq_A$  nor the relation  $\subset_A$  are total orderings of  $A$ .

# Hasse Diagrams

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## Example (informal)

Let  $A = \{a, b, c\}$ . A linearly ordered set  $(\mathcal{P}(A), \preceq)$  represented by your Hasse diagram.

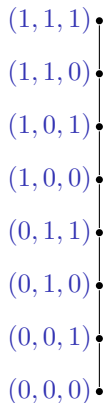


# Hasse Diagrams

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## Example (informal)

Let  $A = \{0, 1\}$ . A linearly ordered set  $(A \times A \times A, \prec)$  represented by your Hasse diagram.



# Outline

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Introduction

Non-Strict Partial Orderings

Strict Partial Orderings

Linear Orderings

**Notable Elements**

References

# Notable Elements

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## Definition [2.]5.11

Let  $\preccurlyeq$  be an ordering of  $P$ , and let  $A \subseteq P$ .

- (a) An element  $a \in A$  is the **least** element of  $A$  in the ordering  $\preccurlyeq$  if  $a \preccurlyeq x$  for every  $x \in A$ .
- (b) An element  $a \in A$  is a **minimal** element of  $A$  in the ordering  $\preccurlyeq$  if there exists no  $x \in A$  such that  $x \preccurlyeq a$  and  $x \neq a$ .

# Notable Elements

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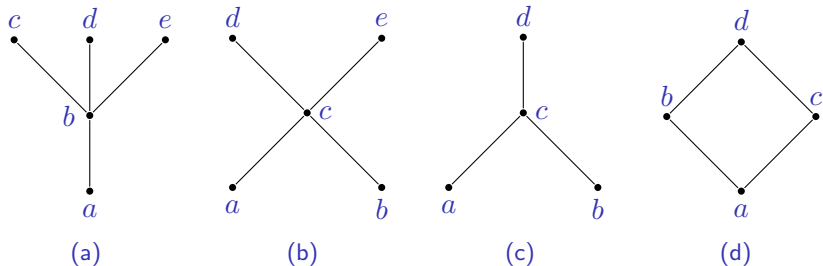
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- (b) An element  $a \in A$  is a **minimal** element of  $A$  in the ordering  $\preccurlyeq$  if there exists no  $x \in A$  such that  $x \preccurlyeq a$  and  $x \neq a$ .

Similarly,

- (a') An element  $a \in A$  is the **greatest** element of  $A$  in the ordering  $\preccurlyeq$  if  $x \preccurlyeq a$  for every  $x \in A$ .
- (b') An element  $a \in A$  is a **maximal** element of  $A$  in the ordering  $\preccurlyeq$  if there exists no  $x \in A$  such that  $a \preccurlyeq x$  and  $x \neq a$ .

# Notable Elements

Example (informal,  $A = P$ )



| Fig. | Least element of $A$ | Greatest element of $A$ | Maximals of $A$ | Minimals of $A$ |
|------|----------------------|-------------------------|-----------------|-----------------|
| (a)  | $a$                  |                         | $c, d, e$       | $a$             |
| (b)  |                      |                         | $d, e$          | $a, b$          |
| (c)  |                      | $d$                     | $d$             | $a, b$          |
| (d)  | $a$                  | $d$                     | $d$             | $a$             |

# Notable Elements

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## Definition [2.]5.14

Let  $\preccurlyeq$  be an ordering of  $P$ , and let  $A \subseteq P$ .

- (a) An element  $p \in P$  is a **lower bound** of  $A$  in the ordered set  $(P, \preccurlyeq)$  if  $p \preccurlyeq x$  for all  $x \in A$ .
- (b) An element  $p \in P$  is the **infimum** (or **greatest lower bound**) of  $A$  in the ordered set  $(P, \preccurlyeq)$ , denoted  $\inf(A)$ , if it is the greatest element of the set of all lower bounds of  $A$  in  $(P, \preccurlyeq)$ .

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Similarly,

- (a') An element  $p \in P$  is an **upper bound** of  $A$  in the ordered set  $(P, \preccurlyeq)$  if  $x \preccurlyeq p$  for all  $x \in A$ .
- (b') An element  $p \in P$  is the **supremum** (or **least upper bound**) of  $A$  in the ordered set  $(P, \preccurlyeq)$ , denoted  $\sup(A)$ , if it is the least element of the set of all upper bounds of  $A$  in  $(P, \preccurlyeq)$ .

# Notable Elements

## Example (informal)

- $A = \{a, b, c\}$

Upper bounds:  $\{e, f, j, h\}$

$\sup(A) = e$

Lower bounds:  $\{a\}$

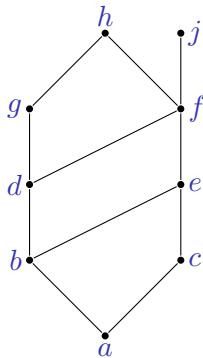
$\inf(A) = e$

- $A = \{j, h\}$

No upper bounds.

Lower bounds:  $\{a, b, c, d, e, f\}$

$\inf(A) = f$



- $A = \{a, c, d, f\}$

Upper bounds:  $\{f, h, j\}$

$\sup(A) = f$

Lower bounds:  $\{a\}$

$\inf(A) = a$

- $A = \{b, d, g\}$

Upper bounds:  $\{g, h\}$

$\sup(A) = g$

Lower bounds:  $\{a, b\}$

$\inf(A) = b$

# Outline

---

Introduction

Non-Strict Partial Orderings

Strict Partial Orderings

Linear Orderings

Notable Elements

References

# References

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Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 4).